Leaf cells

Compaction starts from leaf geometries , cells and wires with axis-aligned boxes

The idea

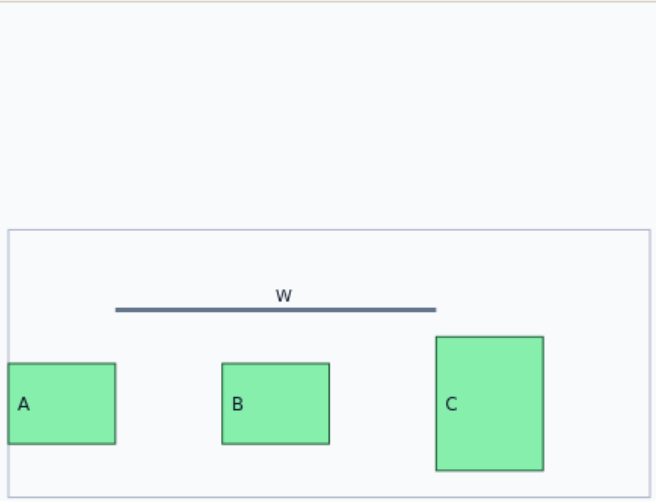
- A,B,C are cells; W is a horizontal wire
- GOLDEN: 4 objects, bbox width 20, height 6

Objects

LEAF CELLS

Objects

Step 1 / 5 · obj · module01-01-leaf-cells



Explanation

Four objects: cells A,B,C and wire W.

- stick layout

```
n=4
```

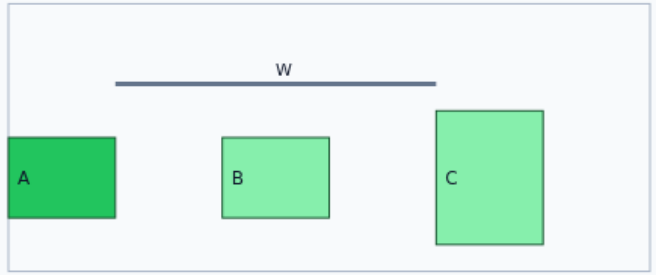
Objects

Cell A

LEAF CELLS

Cell A

Step 2 / 5 · a · module01-01-leaf-cells



Explanation

A sits at x0=0..4, y0=2..5.

- cell

```
left
```

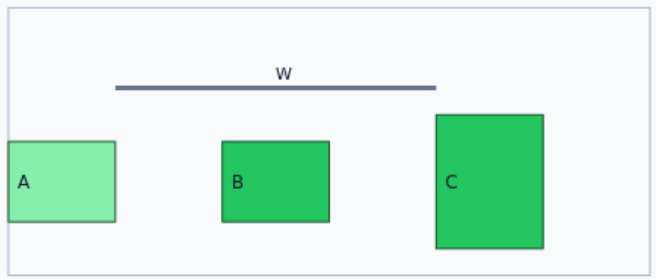
Cell A

B and C

LEAF CELLS

B and C

Step 3 / 5 · bc · module01-01-leaf-cells



Explanation

B and C sit further right with gaps.

- whitespace

```
gaps
```

B and C

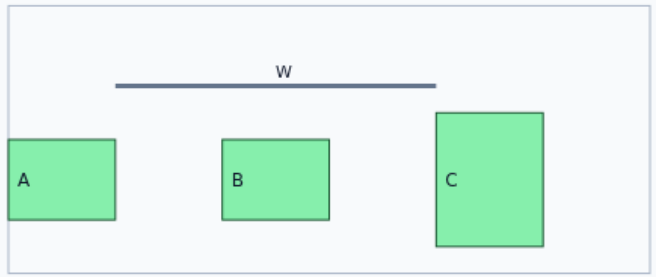
BBox

LEAF CELLS

BBox

Step 4 / 5 · bbox · module01-01-leaf-cells

{"bbox":{"x0":0,"y0":1,"x1":20,"y1":6},"width":20,"height":5,"c



Explanation

Union bbox width=20 height=6.

- bbox

```
W=20
```

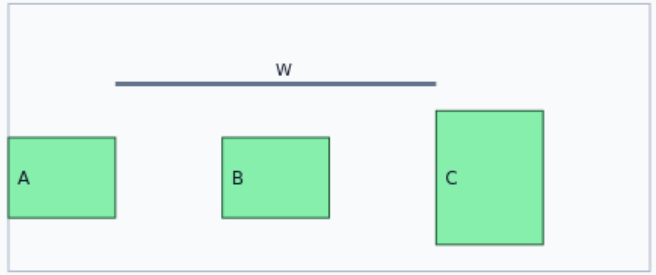
BBox

Count

LEAF CELLS

Count

Step 5 / 5 · count · module01-01-leaf-cells



Explanation

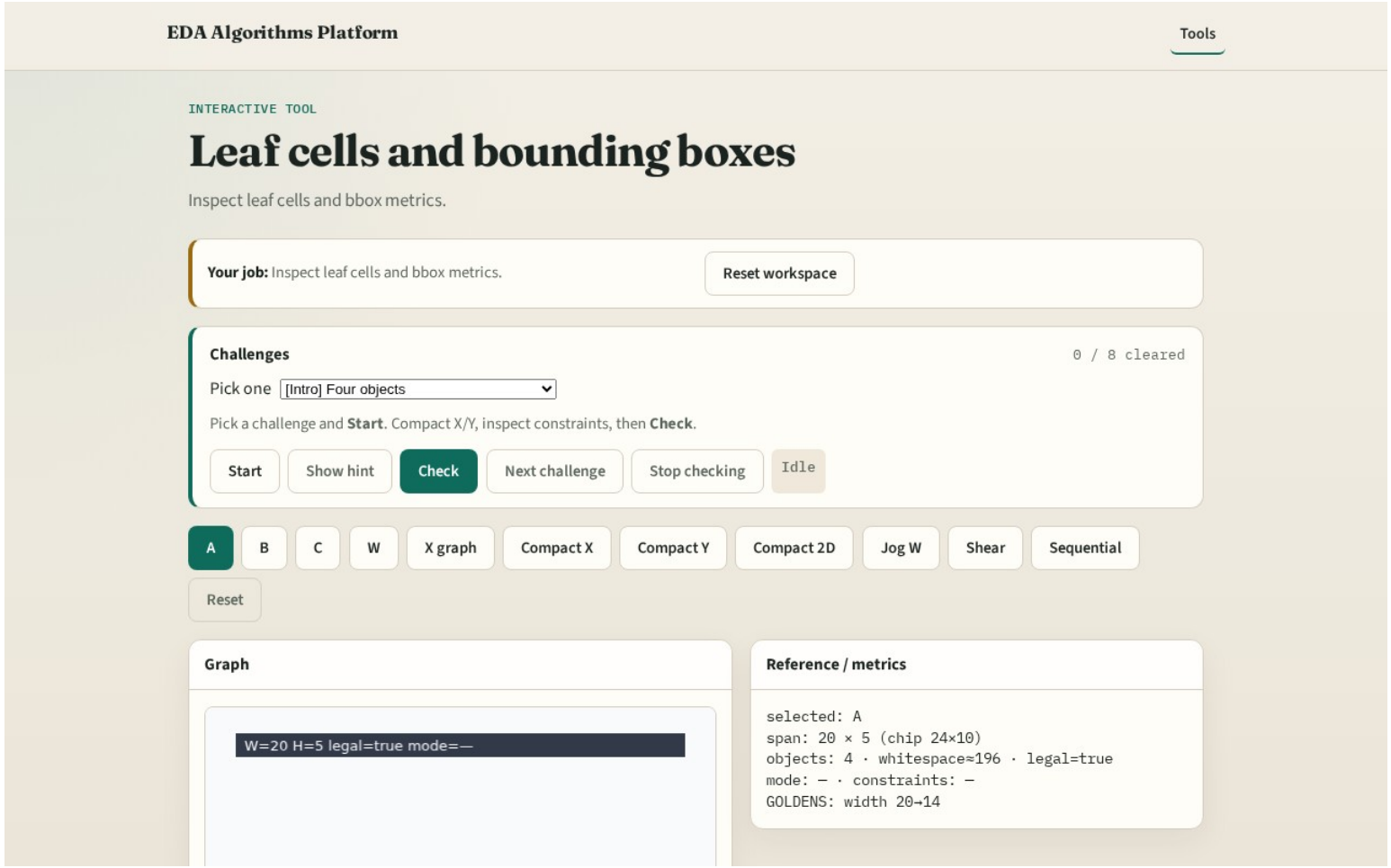
GOLDENS: object_count=4.

- stats

```
4
```

Count

Browser lab track



Browser lab starter

Implement track

- Call layout_stats; assert object_count==4 and width==20

Pitfalls

- Treating wire as a cell
- Confusing chip size with bbox

Your turn

- Next: spacing / gap rules